PSETs Landing Page*

Anish Krishna Lakkapragada

This is the documentation for using my PSET PDFs responsibly. I post these LaTEX'd PSETs (1) as an education resource for friends at other universities, fellow Yalies, and all those interested and (2) for quick reference. These PSETs are not to be used irresponsibly; only look at the solution after giving each problem an honest attempt. **If YOU USE THESE PSETS TO CHEAT, YOU ARE NOT ONLY STUPID BUT YOU ARE CHEATING YOURSELF OUT OF THE ABILITY TO FALL IN LOVE WITH MATH.** Furthermore, I am not smarter than you and my solutions did not always get a perfect score.

The general format for accessing the (one-indexed) `N`th assigned PSET PDF of a Yale course with course number `CODE` is:

`https://anish.lakkapragada.com/notes/TYPE-CODE/psets/N.pdf`

where `TYPE` is `stats` or `math`. Similarily, to access my solution for this PSET you can go to:

`https://anish.lakkapragada.com/notes/TYPE-CODE/sols/N.pdf`

These PSETs and associated solution PDFs are synchronized daily at 4:20AM with my computer files through a Cronjob Shell Script. If you want to contribute any corrections, please email `anish.lakkapragada@yale.edu`.

*Note that PDF here is referring to Portable Document Format, not to be confused with the veritable Probability Density Function.

MATH 226 - HW 1

Anish Lakkapragada – anish.lakkapragada@yale.edu – al2778

September 5, 2024

1.

(a)

Given that image b is the output of $f(a)$ where pre-image $a \in A$, $g(b)$ will always equal a as it is guaranteed that there exists a pre-image a which by function f will map to b. Thus, for all $a \in A$, $g \circ f(a) = a$ and so g is a left inverse of f.

(b)

For g to be the right inverse of f, $f \circ g(b) = b$ where we assume $b \in B$. Because f is an injective function, it is not guaranteed that every image in set B has a corresponding pre-image is in set A as defined by function f. If the given aforementioned b does not have a pre-image in set A as defined by function f, $g(b) = a_0$. And because it is not guaranteed $f(a_0)$ equals b, it is not guaranteed $f(g(b)) = b$. Thus g may not be the right inverse of f.

g would be the right inverse of f is there was a one-to-one correspondence between each pre-image in A and image B through function f. This would occur if f was a bijective function.

(c)

Given the surjective function $f : A \to B$, the right inverse is given by $g(b)$ where $f \circ g(b) = b$. The set of pre-images in A that map by function f to image $b \in B$ is given by the set $P = \{x | f(x) = b\}$. Because f is surjective, $|P| \geq 1$. In cases where $|P| > 1$, $g(b)$ must be able to choose one pre-image from P. I define $g(b)$ as taking any arbitrary pre-image from set P. Because f is surjective, $g(b)$ is guaranteed to return a pre-image that maps by function f to b. Thus the right inverse condition $f \circ g(b) = b$ is upheld, and so $g(b)$ is proved as the right inverse of f.

2.

(a)

We are given that the composition of g with f, $g \circ f$, maps set A to C. This composition $g \circ f$ is injective if any image in C has at most one pre-image in set A. If we assume f is an injective function, we know that only certain pre-images in set A will map to unique elements in set B. Similarily, if we assume g is an injective function, we know that only certain pre-images in set B will map to unique elements

in set C. Because we know a subset of pre-images in A will map to a subset of unique images in B, and a subset of those images (now pre-images) in B will map to unique images in C, we know the composition function $g \circ f$ will map a subset of the pre-images in A to unique images in C. By definition, $g \circ f$ is an injective function if f and g are both injective.

(b)

If we know g is a surjective function, we know that for each image in C, there exists at least one pre-image in B. Similarily, if we know that f is a surjective function, we know that for each of these pre-images that exist in B that map to images in C, there exists at least one pre-image in A. Because every element in C is guaranteed to have at least one pre-image in A by the function $g \circ f$, by defintion, $g \circ f$ is a surjection if f and g are both surjective.

(c)

If functions f and g are both bijective, both functions map all pre-images in A and B respectively to unique images in B and C respectively. Because f provides a one-to-one correspondence between elements in A and B and g provides a one-to-one correspondence between elements in B and C, a one-to-one correspondance is maintained between each pre-image in A and image in C through the function $g(f(a))$ or $g \circ f$. Thus $g \circ f$ is a bijective function if f and g are both bijective.

3.

(a)

Let us define $x = m_1 + n_1\sqrt{2}$, where $m_1, n_1 \in \mathbb{Z}$ and $y = m_2 + n_2\sqrt{2}$, where $m_2, n_2 \in \mathbb{Z}$. Given these definitions, $x + y = (m_1 + m_2) + (n_1 + n_2)\sqrt{2}$. Because $(m_1 + m_2), (n_1 + n_2) \in \mathbb{Z} \Rightarrow x + y \in B$ if $x, y \in B$.

(b)

Let us define x and y the same as in the above part (a). Given these definitions, $xy = m_1m_2 + m_1n_2\sqrt{2} + m_2n_1\sqrt{2} + 2n_1n_2 = (2n_1n_2 + m_1m_2) + (m_1n_2 + m_2n_1)\sqrt{2}$. Because $(2n_1n_2 + m_1m_2), (m_1n_2 + m_2n_1) \in \mathbb{Z} \Rightarrow xy \in B$ if $x, y \in B$.

(c)

For the base case $k = 1$, $(-1 + \sqrt{2})^k = (-1 + \sqrt{2}) \in B$. Given integer $k \geq 1$ and $(-1 + \sqrt{2})^k \in B$, $(-1 + \sqrt{2})^{k+1} \in B$. This is because $(-1 + \sqrt{2})^{k+1} = (-1 + \sqrt{2})^k * (-1 + \sqrt{2})$ and both factors $(-1 + \sqrt{2})^k, (-1 + \sqrt{2}) \in B$. As proven in (b), B is closed under multiplication and so when both factors are in B, their product will be in B. Thus, as proven by induction, for all integers $k \geq 1$, $(-1 + \sqrt{2})^k \in B$.

4.

(a)

Note that for any set C with N items, $T(\Sigma_{i=1}^{N} C_i)$ will equal $\Sigma_{i=1}^{N} T(C_i)$ due to T being an additive function.

For all given integers $n \geq 1$:

$$T(\Sigma_{i=1}^{n} x) = \Sigma_{i=1}^{n} T(x)$$
$$T(nx) = nT(x)$$

(b)

$$T(x+y) = T(x) + T(y)$$
$$T(0+0) = T(0) + T(0)$$
$$T(0) = 2T(0)$$
$$T(0) = 0$$

(c)

$$T(x+y) = T(x) + T(y)$$
$$T(x+(-x)) = T(x) + T(-x)$$
$$T(0) = T(x) + T(-x)$$
$$0 = T(x) + T(-x)$$
$$T(x) = -T(-x)$$

(d) For all integers n and all integers $k \neq 0$,

$$T((\frac{n}{k} * k)x) = \Sigma_{i=1}^{k} T(\frac{n}{k}x) = kT(\frac{n}{k}x)$$

If we define r to be the fraction $\frac{n}{k}$, by definition $r \in \mathbb{Q}$ as n and k are both integers where denominator $k \neq 0$. Using r, we can simplify the above expression further:

$$T(nx) = \frac{n}{r}T(rx)$$
$$rT(nx) = nT(rx)$$

As n is defined as $n \in \mathbb{Z}$, we first generalize our proof in (a) that $T(nx) = nT(x)$ from all integers $n \geq 1$ to $n \in \mathbb{Z}$. Given an integer $n < 0$, if $u = nx$, we can use our proof in (c) that for $u \in \mathbb{R}$, $T(u) = -T(-u)$ and thus since $T(-u) = T(|n|x) = |n|T(x)$, we can conclude that in cases where $n < 0$, $T(u) = T(nx) = -|n|T(x) = nT(x)$ or more simply, $T(nx) = nT(x)$. And in cases where $n = 0$, $T(nx) = nT(x)$ as $T(0) = 0$ as proven in (b). Thus our proof in (a) is generalized to $n \in \mathbb{Z}$. Using this result, we can continue simplifying our above expressions.

$$r(nT(x)) = nT(rx)$$
$$rT(x) = T(rx)$$
$$T(rx) = rT(x)$$

for all rational numbers $r \in \mathbb{Q}$.

(e)

Let us define $T(x)$:

$$T(x) = \begin{cases} x, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \notin \mathbb{Q} \end{cases}$$

Defining $r = \sqrt{2}, x = 1$:

$$\begin{gathered} T(\sqrt{2}x) = \sqrt{2}T(x) \\ T(\sqrt{2}) = \sqrt{2} * T(1) \\ 0 \neq 1 \\ T(rx) \neq rT(x) \end{gathered}$$

As proven by contradiction, $T(rx) \neq rT(x)$ for all reals $r \in \mathbb{R}$.

5.

(a)

Let us define field $\mathbb{F} = (\mathbb{Z}\{\sqrt{3}\}, +, \cdot)$ and the multiplicative inverse of $a + b\sqrt{3}$ as z where $(a + b\sqrt{3})z = 1$. Given that $a^2 - 3b^2 \neq 0$, z is given by $\frac{1}{a+b\sqrt{3}} = \frac{a-b\sqrt{3}}{a^2-3b^2}$. In the case where $a^2 - 3b^2 = 1$, $z = a - b\sqrt{3}$. Because $a - b\sqrt{3} \in \mathbb{F}$, $a + b\sqrt{3}$ has a multiplicative inverse in this case. Similarily, in the case where $a^2 - 3b^2 = -1$, $z = -a + b\sqrt{3}$. Because $-a + b\sqrt{3} \in \mathbb{F}$, $a + b\sqrt{3}$ has a multiplicative inverse in this case as well.

(b)

Given that $a + b\sqrt{3} \in \mathbb{F}$ has a multiplicative inverse, let us define this multiplicative inverse as $c + d\sqrt{3} \in \mathbb{F}$ where $c, d \in \mathbb{Z}$ and $(a + b\sqrt{3})(c + d\sqrt{3}) = 1$. Let us also define the greatest common divisor of a and b as $k = gcd(a, b) \in \mathbb{Z}$ where $a = ka'$ and $b = kb'$ and $a', b' \in \mathbb{Z}$ are coprime. We inspect the possible values of k below.

$$\begin{gathered} (a + b\sqrt{3}) * (c + d\sqrt{3}) = 1 \\ (ka' + kb'\sqrt{3}) * (c + d\sqrt{3}) = 1 \\ ka'c + ka'd\sqrt{3} + kb'c\sqrt{3} + 3kb'd = 1 + 0\sqrt{3} \\ ka'c + 3kb'd = 1 \\ k(a'c + 3b'd) = 1 \\ (a'c + 3b'd) = \frac{1}{k} \end{gathered}$$

Because $a'c + 3b'd \in \mathbb{Z} \Rightarrow k \leq 1$. Because $k \in \mathbb{Z} \Rightarrow k = 1$. We now define the values of c, d in terms of a, b. Defining $z = c + d\sqrt{3}$, $z = \frac{1}{a+b\sqrt{3}} = \frac{a-b\sqrt{3}}{a^2-3b^2} \Rightarrow c = \frac{a}{a^2-3b^2}, d = \frac{-b}{a^2-3b^2}$. Because the largest possible value that can divide two integers, a, b, into integers is given by $k = gcd(a, b)$, the denominator in c, d of $a^2 - 3b^2$ must equal $k = 1$ in order to ensure $c, d \in \mathbb{Z}$ so that $c + d\sqrt{3} \in \mathbb{F}$. Note that $a^2 - 3b^2 = -1$ also ensures the multiplicative inverse $c + d\sqrt{3} \in \mathbb{F}$ as $c, d \in \mathbb{Z}$ because $\pm a, \pm b \in \mathbb{Z}$. Thus, if $a + b\sqrt{3} \in \mathbb{F}$ has a multiplicative inverse, we know that $|a^2 - 3b^2| = 1$. If there is no multiplicative inverse in $\mathbb{F}$, that is because $c \notin \mathbb{Z}$ or $d \notin \mathbb{Z}$, which would happen only if the denominator $|a^2 - 3b^2| \neq gcd(a, b)$ or $|a^2 - 3b^2| \neq 1$. Thus, if $a + b\sqrt{3} \in \mathbb{F}$ has a multiplicative inverse $\iff |a^2 - 3b^2| = 1$.

(c)

In order for $\mathbb{F} = (\mathbb{Z}\{\sqrt{3}\}, +, \cdot)$ to define a field, $\forall m \in \mathbb{F}$, $\exists n \in \mathbb{F}$ such that $m \cdot n = 1$. However, as shown in (b), $a + b\sqrt{3} \in \mathbb{F}$ will only have a guaranteed multiplicative inverse in the special case that $|a^2 - 3b^2| = 1$. Thus, $\mathbb{F}$ fails to meet the muliplicative inverse condition to be defined as a valid field.